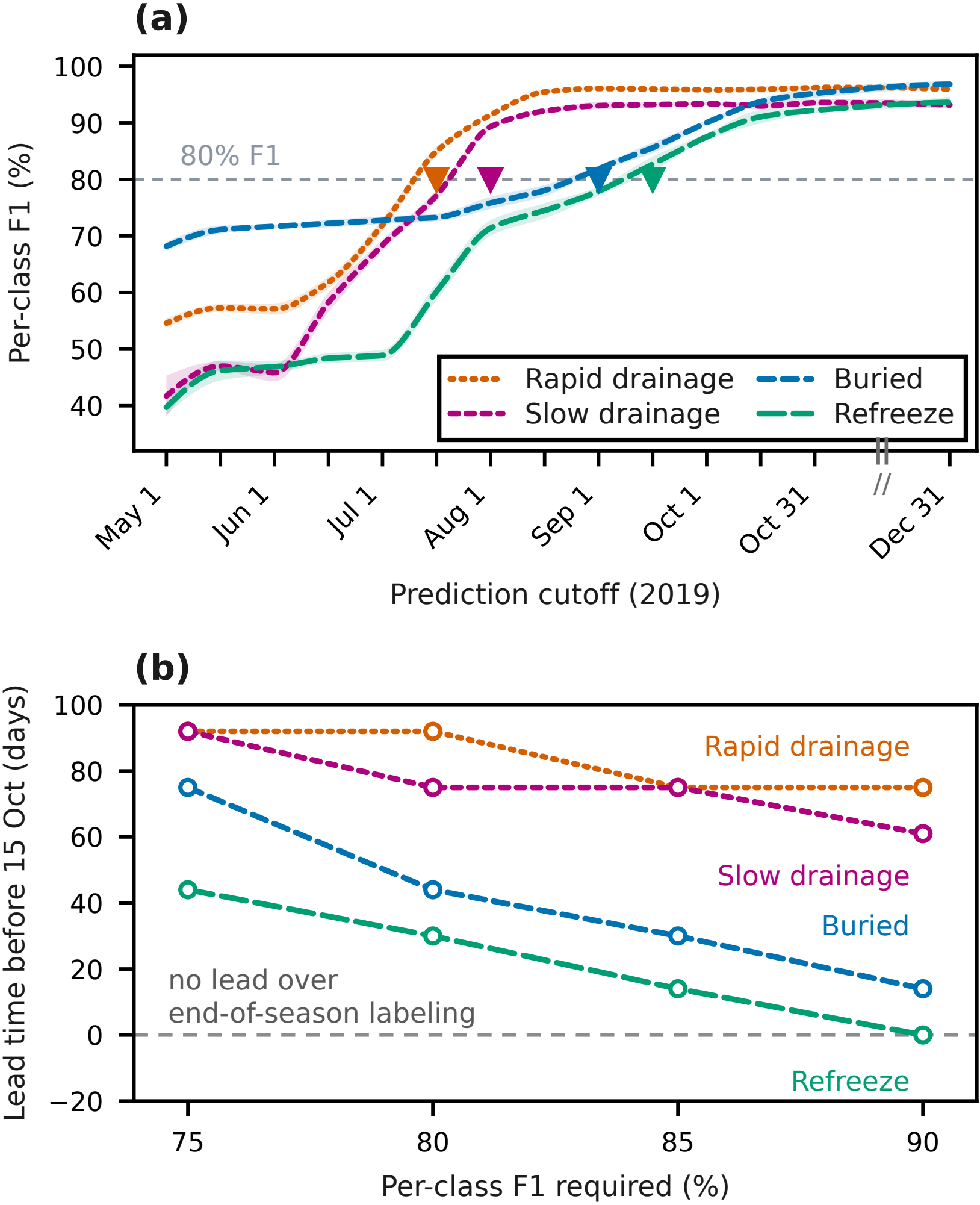
(a)
100
90
80
70
60
50
40
Per-class F1 (%)
80% F1
Rapid drainage
Slow drainage
Buried
Refreeze
May 1
Jun 1
Jul 1
Aug 1
Sep 1
Oct 1
Oct 31
Dec 31
Prediction cutoff (2019)
(b)
100
80
60
40
20
0
−20
Lead time before 15 Oct (days)
Rapid drainage
Slow drainage
Buried
Refreeze
no lead over
end-of-season labeling
75
80
85
90
Per-class F1 required (%)